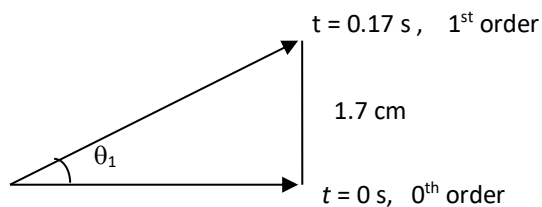


7	a	When the wavelength of the wave is comparable to slit width, diffraction is most significant.	1
			1
	b	i Distance D must be much longer than distance between two adjacent slits of the grating	1
		ii Slit separation, $d = 1/p$	1
		path difference $x = d \sin \theta = 1/p \sin \theta$	1
		iii Hence constructive interference occurs when	
		$1/p \sin \theta = n \lambda$ where $n = 0, 1, 2, \dots$	1
		Or $\sin \theta = n p \lambda$ where $n = 0, 1, 2, \dots$	
		Where n is the order of the spectrum and θ is the angle of diffraction between the diffracted ray and the direction of incident light.	1
	c	i Angular speed of grating, $\omega = 2 \pi / T = 2 \pi / 3.0$ $= 2.09 = 2.1 \text{ rad s}^{-1}$	1
			1
		ii Peak C corresponds to the zeroth order of the interference pattern whereas peaks B and D the first order, and Peaks A and E the second order.	1
		The higher the order of the peaks, the lower the height of each peak due to the longer distance that it has to travel to reach the screen.	1
		iii Time taken to shift from zeroth order to the first order through 1.7 cm on the CRO screen is $1.7 \times 0.10 \text{ s} = 0.17 \text{ s}$	1



$$\theta_1 = (2 \pi / 3) \times 0.17 [1] = 0.356 \text{ rad} = 0.36 \text{ rad} \quad 1$$

Similarly time from the zeroth order to 2nd order = $3.7 \times 0.1 = 0.37 \text{ s}$

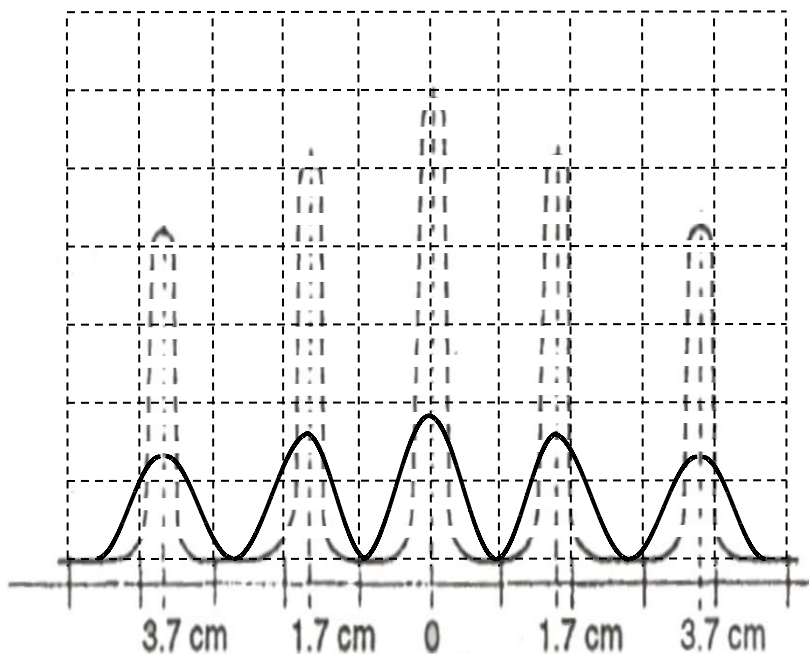
$$\theta = (2 \pi / 3) \times 0.37 = 0.7749 \text{ rad} = 0.77 \text{ rad}. \quad 1$$

IV Using the grating equation

$$\sin \theta_2 = 2 \times 5.5 \times 10^5 \times \lambda \quad 1$$

$$\lambda = 636 \text{ nm} \quad 1$$

V Peak E is preferred as the angle θ is larger and the percentage error for calculating the wavelength is smaller. 1



Fringe separation remains the same. [1]

Lower intensity for all peaks [1]

vii Since the slit separation remain the same, the fringe separation remains the same. 1

However, the peaks are less intense (poorer contrast) and less sharp (broader and less defined) due to fewer slits that allow light to pass through. 1